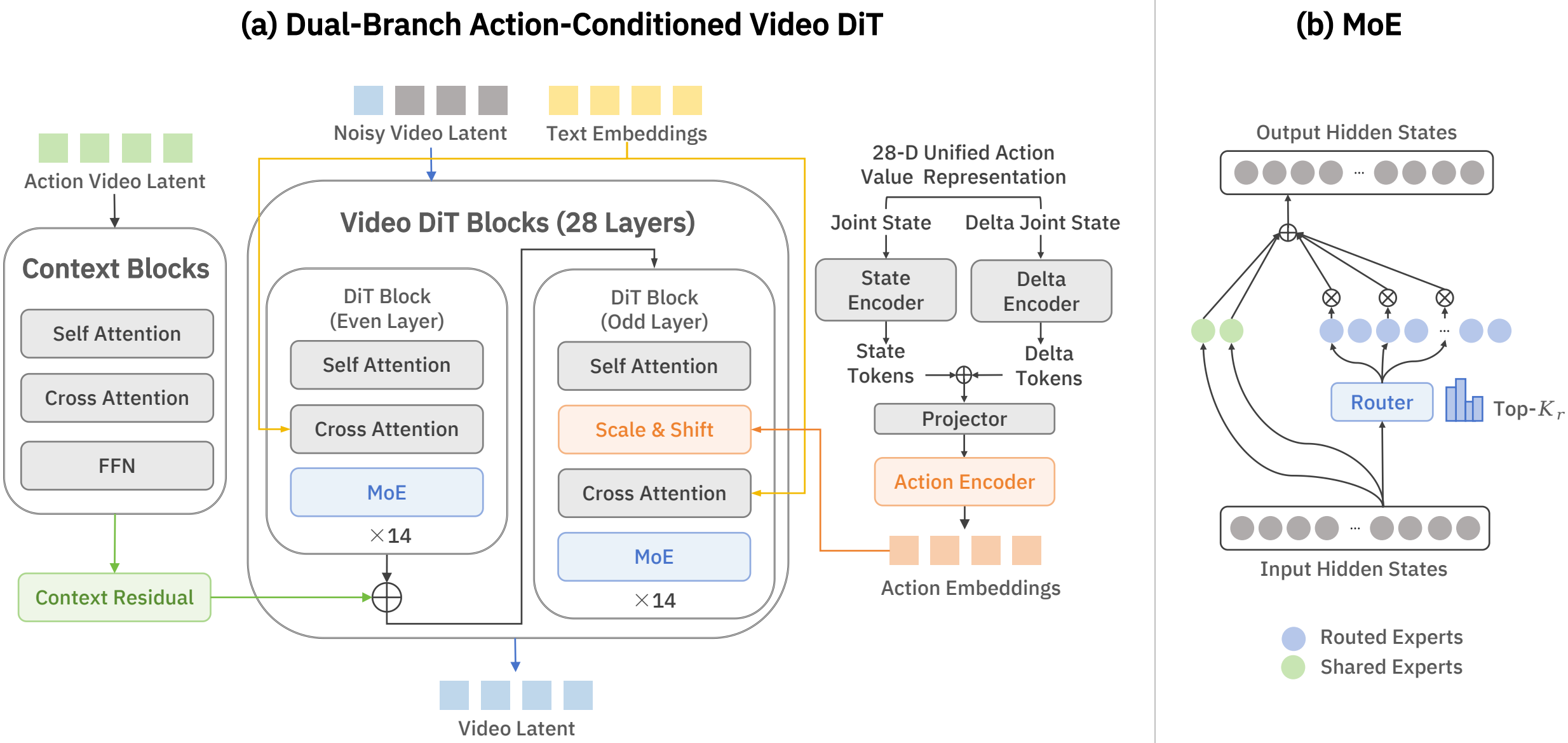
(a) Dual-Branch Action-Conditioned Video DiT
Noisy Video Latent
Text Embeddings
Action Video Latent
Context Blocks
Self Attention
Cross Attention
FFN
Context Residual
Video DiT Blocks (28 Layers)
DiT Block (Even Layer)
Self Attention
Cross Attention
MoE
×14
DiT Block (Odd Layer)
Self Attention
Scale & Shift
Cross Attention
MoE
×14
Video Latent
28-D Unified Action Value Representation
Joint State
Delta Joint State
State Encoder
Delta Encoder
State Tokens
Delta Tokens
Projector
Action Encoder
Action Embeddings
(b) MoE
Output Hidden States
Router
Top-K_r
Input Hidden States
Routed Experts
Shared Experts